
10 / Filter Design by Modeling

10.0 / Introduction

The two preceding chapters presented the traditional design methods for IIR and FIR digital filters. These methods are grounded in the mathematical theory of approximation, where upper and lower constraints are placed on the deviation of $|H'(\omega)|$ from the ideal response. Another class of design methods is based on the statistical modeling of spectra; and in these methods, the least-squares error (LSE) criterion is employed. These methods are especially appropriate when the filter design or model is derived from experimental data having random fluctuations. However, they can also be useful for deterministic designs when minimum-phase FIR filters are desired, when nonclassical specifications are given for IIR filters, when poles and zeros of unequal number ($N \neq M$) are desired, and so forth. Complete coverage of this field, which is sometimes termed *modern spectrum analysis*, is beyond the scope of this book, and we will concentrate on deterministic filter design. However, the basics of statistical data modeling and spectrum analysis will also be included in section 10.5.

Common to all of the modeling approaches to be covered in this chapter is the need for approximate solutions to overdetermined sets of linear simultaneous equations, i.e., where there are more equations than unknowns. This is a classical problem in linear algebra, and a simple, elegant theory exists for its solution based upon the least-squares minimization of the associated equation errors [19]. We thus begin with a brief introduction to (review of) this technique using convenient matrix notation.

Overdetermined Equations

Consider a set of linear simultaneous equations of the form

$$\mathbf{F}\mathbf{x} = \mathbf{g} \quad (10.0.1)$$

where $\mathbf{F}$ is a known $L \times N$ matrix (L rows by N columns), $\mathbf{x}$ is an N -element vector of unknowns, and $\mathbf{g}$ is a known L -element vector. If $L = N$, the matrix $\mathbf{F}$ is square, and there are as many equations as unknowns. If $\mathbf{F}$ is also nonsingular, a unique solution $\mathbf{x}$ exists and is given by

$$\mathbf{x} = \mathbf{F}^{-1}\mathbf{g} \quad (10.0.2)$$

where $\mathbf{F}^{-1}$ is the inverse matrix satisfying $\mathbf{F}^{-1}\mathbf{F} = \mathbf{I}$. If $L > N$, however, $\mathbf{F}$ is a nonsquare matrix having more rows than columns, and there are more equations in (10.0.1) than unknowns. In this case, the equations are inconsistent, in general, and a solution $\mathbf{x}$ does not exist unless the vector $\mathbf{g}$ happens to lie in the N -dimensional subspace spanned by the columns of $\mathbf{F}$.

Therefore, when $L > N$ we must modify (10.0.1), in general, to obtain a solution $\mathbf{x}$ by adding an error vector $\mathbf{e}$, i.e.,

$$\mathbf{F}\mathbf{x} = \mathbf{g} + \mathbf{e}. \quad (10.0.3)$$

We then choose an appropriate and tractable error criterion and seek the solution $\mathbf{x}$ that satisfies this criterion. The most common criterion is the LSE criterion that minimizes the sum of the squared errors from the L equations, i.e.,

$$E = \mathbf{e}'\mathbf{e} = \sum_{i=1}^L e_i^2. \quad (10.0.4)$$

(Note that $\mathbf{e}'\mathbf{e}$ is simply the inner (dot) product of $\mathbf{e}$ with itself.) Writing $\mathbf{e}$ as $(\mathbf{F}\mathbf{x} - \mathbf{g})$ from (10.0.3), we then have

$$\begin{aligned} E &= (\mathbf{F}\mathbf{x} - \mathbf{g})'(\mathbf{F}\mathbf{x} - \mathbf{g}) \\ &= (\mathbf{x}'\mathbf{F}' - \mathbf{g}')(\mathbf{F}\mathbf{x} - \mathbf{g}) \\ &= \mathbf{x}'\mathbf{F}'\mathbf{F}\mathbf{x} - 2\mathbf{x}'\mathbf{F}'\mathbf{g} + \mathbf{g}'\mathbf{g}, \end{aligned} \quad (10.0.5)$$

because $\mathbf{g}'\mathbf{F}\mathbf{x} = \mathbf{x}'\mathbf{F}'\mathbf{g}$. But if E is to be minimized by our choice of $\mathbf{x}$, we must have

$$\frac{\partial E}{\partial x_i} = 0, \quad i = 1, 2, \dots, N,$$

or in vector notation,

$$\frac{\partial E}{\partial \underline{\mathbf{x}}} = \underline{\mathbf{0}}. \quad (10.0.6)$$

The substitution of (10.0.5) into (10.0.6) produces

$$2\mathbf{F}'\mathbf{F}\underline{\mathbf{x}} - 2\mathbf{F}'\mathbf{g} = \underline{\mathbf{0}}$$

or simply

$$\mathbf{F}'\mathbf{F}\underline{\mathbf{x}} = \mathbf{F}'\mathbf{g}. \quad (10.0.7)$$

Equation (10.0.7) is the desired *normal* equation for the LSE solution $\underline{\mathbf{x}}$. (If the reader is unfamiliar with vector differentiation, the same result can be obtained by writing (10.0.5) in terms of summations and taking the individual partial derivatives with respect to the x_i to produce the N equations in (10.0.7).)

Note that if both sides of (10.0.3) are premultiplied by $\mathbf{F}'$, we have

$$\mathbf{F}'\mathbf{F}\underline{\mathbf{x}} = \mathbf{F}'\mathbf{g} + \mathbf{F}'\underline{\mathbf{e}}. \quad (10.0.8)$$

Comparing (10.0.7) with (10.0.8), we find that

$$\mathbf{F}'\underline{\mathbf{e}} = \underline{\mathbf{0}}. \quad (10.0.9)$$

This is an example of the famous *orthogonality principle* of least-squares estimation, which states that the error vector $\underline{\mathbf{e}}$ corresponding to the minimizing solution $\underline{\mathbf{x}}$ must be orthogonal to each of the columns of $\mathbf{F}$ and thus to the subspace spanned by the columns of $\mathbf{F}$ [20].

If the $N \times N$ matrix $\mathbf{F}'\mathbf{F}$ is nonsingular, the solution to the normal equation in (10.0.7) is unique and is simply

$$\underline{\mathbf{x}} = (\mathbf{F}'\mathbf{F})^{-1}\mathbf{F}'\mathbf{g} \quad \text{or} \quad \underline{\mathbf{x}} = \mathbf{F}^\# \mathbf{g} \quad (10.0.10)$$

where $\mathbf{F}^\# = (\mathbf{F}'\mathbf{F})^{-1}\mathbf{F}'$ is the *pseudo-inverse* of $\mathbf{F}$ (also called the *generalized inverse* or the *Moore-Penrose inverse*). $\mathbf{F}'\mathbf{F}$ is nonsingu-

lar if, and only if, the rank of $\mathbf{F}$ is N , i.e., if the N columns of $\mathbf{F}$ are linearly independent. On the other hand, if $\mathbf{F}'\mathbf{F}$ is singular, a solution $\mathbf{x}$ still exists but is nonunique. In that case, the pseudo-inverse cannot be defined as in (10.0.10), but instead is defined in terms of the singular-value decomposition of $\mathbf{F}$ [19].

10.1 / Autoregressive (All-pole) Filters

The techniques of *autoregressive* (AR) or all-pole filter modeling underlie all of the methods in this field. Consider the causal AR filter model

$$H(z) = \frac{1}{A(z)} \quad (10.1.1)$$

where

$$A(z) = a_0 + a_1 z^{-1} + \cdots + a_N z^{-N}.$$

Then

$$H(z)A(z) = 1 \quad (10.1.2)$$

or, in the time domain, using the alternate notation $a(n) = a_n$,

$$h(n) * a(n) = \delta(n). \quad (10.1.3)$$

The FIR filter $A(z)$ is thus the *inverse filter* for $H(z)$, which *whitens* $h(n)$ to produce $\delta(n)$, as depicted in figure 10.1(a). Using the $2N+1$ consecutive values for $h(n)$ beginning at $n = -N$, (10.1.3) provides $N+1$ linear equations from which to solve for $a(n)$. That is, given $h(0)$ through $h(N)$ plus the fact that $h(n) = 0$, $n < 0$, the convolution may be written in matrix form to produce the following $N+1$ equations for $a(n)$:

$$\begin{bmatrix} h_0 & 0 & \cdot & \cdot & \cdot & 0 \\ h_1 & h_0 & \cdot & \cdot & \cdot & 0 \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ h_N & \cdot & \cdot & \cdot & \cdot & h_0 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ \cdot \\ \cdot \\ \cdot \\ a_N \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ \cdot \\ \cdot \\ \cdot \\ 0 \end{bmatrix}. \quad (10.1.4)$$

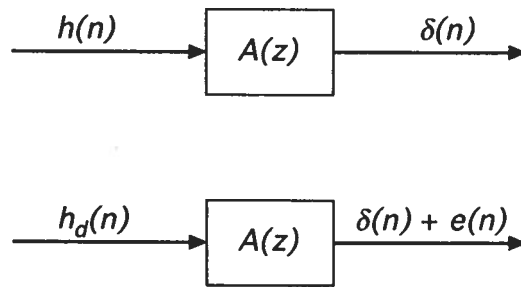


Figure 10.1. Inverse filter $A(z)$ applied to (a) model response $h(n)$ and (b) desired response or data $h_d(n)$.

However, given a desired or measured impulse response $h_d(n)$ that is not exactly AR and/or of greater than N th order, we can only approximate it as in (10.1.1). Therefore, (10.1.3) becomes

$$h_d(n) * a(n) = \delta(n) + e(n), \quad (10.1.5)$$

where $e(n)$ is the approximation error, and $A(z)$ is only an approximate inverse filter as illustrated in figure 10.1(b). Writing (10.1.5) in matrix form, we have

$$\mathbf{H}_d \mathbf{a} = \underline{\delta} + \underline{e} \quad (10.1.6)$$

where

$$\mathbf{H}_d = \begin{bmatrix} h_{d0} & 0 & \cdot & \cdot & \cdot & 0 \\ h_{d1} & h_{d0} & & & & 0 \\ \cdot & & \cdot & & & \cdot \\ \cdot & & & \cdot & & \cdot \\ \cdot & & & & \cdot & \cdot \\ h_{dN} & h_{d,N-1} & \cdot & \cdot & \cdot & h_{d0} \\ \cdot & & & & & \cdot \\ \cdot & & & & & \cdot \\ \cdot & & & & & \cdot \\ h_{dL} & \cdot & \cdot & \cdot & \cdot & h_{d,L-N} \end{bmatrix},$$

$$\underline{a} = [a_0, a_1, \dots, a_N]^t,$$

$$\underline{\delta} = [1, 0, 0, \dots, 0]^t,$$

$$\underline{e} = [e_0, e_1, \dots, e_L]^t.$$

Note that we have assumed more data samples $h_d(n)$, $n = 0, 1, \dots, L$, than the minimum required (i.e., $L > N$) in order to approximate $h_d(n)$ closely (hopefully) over more than the minimum interval. Applying the least-squares error criterion, we seek to minimize

$$E_N = \underline{e}'\underline{e} = \sum_{n=0}^L e^2(n). \quad (10.1.7)$$

This is the standard problem in least-squares estimation with over-determined equations [19], discussed in section 10.0. The minimizing solution $\underline{a}$ is obtained by multiplying through (10.1.6) by $\mathbf{H}_d'$ to produce

$$\mathbf{H}_d' \mathbf{H}_d \underline{a} = \mathbf{H}_d' \underline{\delta} + \mathbf{H}_d' \underline{e}. \quad (10.1.8)$$

It is then found as shown earlier in (10.0.9) that

$$\mathbf{H}_d' \underline{e} = \underline{0}. \quad (10.1.9)$$

Note also that $\mathbf{H}_d' \underline{\delta} = h_{d0} \underline{\delta}$, where the dimension of $\underline{\delta}$ varies as required.

Covariance Method

In more compact notation, therefore, we have from (10.1.8) and (10.1.9) the normal equation

$$\Phi_N \underline{a} = h_{d0} \underline{\delta} \quad (10.1.10)$$

where Φ_N is the $(N+1) \times (N+1)$ symmetric *covariance* matrix

$$\Phi_N = \mathbf{H}_d' \mathbf{H}_d \quad (10.1.11)$$

and thus

$$\underline{a} = h_{d0} \Phi_N^{-1} \underline{\delta}. \quad (10.1.12)$$

That is, $\underline{a}$ is proportional to the first column of Φ_N^{-1} . Φ_N is a positive-definite matrix and thus invertible because the columns of $\mathbf{H}_d$ are linearly independent. This general approach to least-squares estimation of AR models (having a finite-duration squared-error norm without windowing, as in (10.1.7)) is known as the *covariance method of linear prediction* [21]. Efficient inversion of Φ_N is achieved by Cholesky factorization.

Yule-Walker Equation

As $L \rightarrow \infty$, the elements ϕ_{ij} of Φ_N approach the autocorrelation values $r(i-j)$ where

$$r(m) = \sum_{n=0}^{\infty} h_d(n)h_d(n+m), \quad (10.1.13)$$

and since $h_d(n)$ is real, $r(-m) = r(m)$. Therefore, Φ_N approaches the symmetric Toeplitz autocorrelation matrix

$$\mathbf{R}_N = \begin{bmatrix} r_0 & r_1 & \cdot & \cdot & \cdot & r_N \\ r_1 & r_0 & & & & \cdot \\ \cdot & & \cdot & & & \cdot \\ \cdot & & & \cdot & & \cdot \\ \cdot & & & & \cdot & \cdot \\ r_N & \cdot & \cdot & \cdot & \cdot & r_0 \end{bmatrix}.$$

The limiting form of (10.1.10) with $\Phi_N = \mathbf{R}_N$ is known as the *Yule-Walker equation*, that is,

$$\mathbf{R}_N \underline{a} = h_{d0} \underline{\delta}. \quad (10.1.14)$$

Additional insight may be gained in this limiting case by transforming (10.1.5) to obtain

$$E(z) = H_d(z)A(z) - 1.$$

By Parseval's Theorem, therefore, the error norm becomes

$$E_N = \sum_{n=0}^{\infty} e^2(n) = \frac{1}{2\pi} \int_0^{2\pi} |H'_d(\omega)A'(\omega) - 1|^2 d\omega. \quad (10.1.15)$$

This shows in the frequency domain the sense in which $A'(\omega)$ whitens $H'_d(\omega)$. Clearly, from (10.1.4), $E_N \rightarrow 0$ as the order $N \rightarrow \infty$.

Hence, any response $H'_d(\omega)$, whether AR or not, may be approximated arbitrarily closely by $H'(\omega) = 1/A'(\omega)$ in the sense of (10.1.15) if N is chosen to be sufficiently large.

Autocorrelation Method

Given only a finite data block $h_d(n)$, $n = 0, 1, \dots, L$, $\mathbf{R}_N$ may be approximated by zero padding the data to produce

$$\hat{\mathbf{R}}_N = \hat{\mathbf{H}}_d^t \hat{\mathbf{H}}_d = \begin{bmatrix} \hat{r}_0 & \hat{r}_1 & \cdot & \cdot & \cdot & \hat{r}_N \\ \hat{r}_1 & \hat{r}_0 & & & & \cdot \\ \cdot & & \cdot & & & \cdot \\ \cdot & & & \cdot & & \cdot \\ \cdot & & & & \cdot & \cdot \\ \hat{r}_N & \cdot & \cdot & \cdot & \cdot & \hat{r}_0 \end{bmatrix}, \quad (10.1.16)$$

where

$$\hat{\mathbf{H}}_d = \begin{bmatrix} h_{d0} & 0 & \cdot & \cdot & \cdot & 0 \\ h_{d1} & h_{d0} & & & & 0 \\ \cdot & \cdot & \cdot & & & \cdot \\ \cdot & & & \cdot & & \cdot \\ \cdot & & & & \cdot & \cdot \\ \cdot & & & & & h_{d0} \\ h_{dL} & & & & & \cdot \\ 0 & \cdot & & & & \cdot \\ \cdot & & \cdot & & & \cdot \\ \cdot & & & \cdot & & \cdot \\ \cdot & & & & \cdot & \cdot \\ 0 & \cdot & \cdot & \cdot & \cdot & h_{dL} \end{bmatrix}$$

and thus

$$\hat{r}(m) = \sum_{n=0}^{L-m} h_d(n) h_d(n+m) \quad (10.1.17)$$

for $0 \leq m \leq N$. Substitution of $\hat{\mathbf{R}}_N$ in place of $\mathbf{R}_N$ in (10.1.14) constitutes the *autocorrelation method of linear prediction* [21]. That is, with the normalization of $\underline{a}$ given by

$$\hat{\underline{a}} = \underline{a}/a_0 = [1, \hat{a}_1, \dots, \hat{a}_N]^t,$$

the autocorrelation method corresponds to the normal equation

$$\hat{\mathbf{R}}_N \hat{\mathbf{a}} = (h_{d0}/a_0) \underline{\delta} \quad (10.1.18)$$

or

$$\hat{\mathbf{a}} = (h_{d0}/a_0) \hat{\mathbf{R}}_N^{-1} \underline{\delta}. \quad (10.1.19)$$

Since the columns of $\hat{\mathbf{H}}_d$ are linearly independent, $\hat{\mathbf{R}}_N$ cannot be singular.

The autocorrelation method, in effect, windows the infinite sequence $h_d(n)$ by a rectangular window $w_R(n)$ of length $L+1$, and hence the desired frequency response $H_d(\omega)$ is convolved with the window transform $W_R(\omega)$. The resulting AR filter is thus an approximation to $H_d(\omega) \otimes W_R(\omega)$ itself. In return for this disadvantage, however, there are several advantages to this method. One is that $A(z)$ is guaranteed to be minimum phase (all roots inside the unit circle), and hence $H(z)$ is guaranteed to be stable. This is not true for the covariance method, in general. In fact, if for some reason one wishes to model an unstable $h_d(n)$, the covariance method is the appropriate choice. However, for a stable $h_d(n)$, the covariance method will usually also produce a stable model. The proof of stability for the autocorrelation method will be given shortly.

EXAMPLE

As a simple example of the covariance and autocorrelation methods, we will obtain first-order models of the desired response $h_d(n) = c^n$, for $n = 0, 1, \dots, L$, by these two methods. For the covariance method, we form the data matrix

$$\mathbf{H}_d = \begin{bmatrix} 1 & 0 \\ c & 1 \\ c^2 & c \\ \cdot & \cdot \\ \cdot & \cdot \\ \cdot & \cdot \\ c^L & c^{L-1} \end{bmatrix}$$

from which the covariance matrix for $N=1$ is seen to be

$$\Phi_1 = \mathbf{H}_d^t \mathbf{H}_d = \begin{bmatrix} \sum_{n=0}^L c^{2n} & \sum_{n=0}^{L-1} c^{2n+1} \\ \sum_{n=0}^{L-1} c^{2n+1} & \sum_{n=0}^{L-1} c^{2n} \end{bmatrix}.$$

Since $h_{d0} = 1$, the normal equation for $\underline{a}$ is of the form

$$\Phi_1 \underline{a} = \underline{\delta}.$$

The two simultaneous linear equations composing this vector equation yield the expected solution

$$\underline{a} = \begin{bmatrix} 1 \\ -c \end{bmatrix}.$$

Therefore, the filter model is just

$$H(z) = \frac{1}{1 - cz^{-1}}$$

or $h(n) = c^n u(n)$, which exactly matches the given $h_d(n)$ in this simple case, whether stable ($|c| < 1$) or not. For the autocorrelation method, we zero-pad the given data to produce

$$\hat{\mathbf{H}}_d = \begin{bmatrix} 1 & 0 \\ c & 1 \\ c^2 & c \\ \vdots & \vdots \\ \vdots & c^2 \\ \vdots & \vdots \\ c^L & \vdots \\ 0 & c^L \end{bmatrix}$$

and thus

$$\hat{\mathbf{R}}_1 = \hat{\mathbf{H}}_d^t \hat{\mathbf{H}}_d = \begin{bmatrix} \sum_{n=0}^L c^{2n} & \sum_{n=0}^{L-1} c^{2n+1} \\ \sum_{n=0}^{L-1} c^{2n+1} & \sum_{n=0}^L c^{2n} \end{bmatrix}.$$

The normal equation is now

$$\hat{\mathbf{R}}_1 \hat{\mathbf{a}} = (1/a_0) \hat{\underline{d}}.$$

Employing the second of the two equations comprising this vector equation to solve for $\hat{a}_1$, we have

$$\sum_{n=0}^{L-1} c^{2n+1} + \hat{a}_1 \sum_{n=0}^L c^{2n} = 0,$$

from which

$$\hat{a}_1 = - \left(\frac{\sum_{n=0}^{L-1} c^{2n+1}}{\sum_{n=0}^L c^{2n}} \right).$$

One can show that $|\hat{a}_1| < 1$ and hence that the model is stable for all values of c . Note, in particular, the following cases:

$$\hat{a}_1 \approx -c, \quad |c| \ll 1$$

$$\hat{a}_1 \approx -1/c, \quad |c| \gg 1$$

$$\hat{a}_1 = -L/(L+1), \quad c = 1.$$

Levinson-Durbin Algorithm

A second advantage of the autocorrelation method in addition to guaranteed stability is that a very efficient iterative algorithm exists for the solution of $\hat{\mathbf{a}}$. This is the Levinson-Durbin algorithm, which solves for $\hat{\mathbf{a}}$ recursively from first to N th order. Let the P th-order solution be $\hat{\mathbf{a}}_P = \underline{a}_P/a_{0P}$, i.e.,

$$\hat{\mathbf{R}}_P \hat{\mathbf{a}}_P = (h_{d0}/a_{0P}) \hat{\underline{d}}. \quad (10.1.20)$$

Assume the recursive relationship

$$\hat{\mathbf{a}}_{P+1} = \begin{bmatrix} 1 \\ \hat{a}_{1P} \\ \cdot \\ \cdot \\ \hat{a}_{PP} \\ 0 \end{bmatrix} + K_{P+1} \begin{bmatrix} 0 \\ \hat{a}_{PP} \\ \cdot \\ \cdot \\ \hat{a}_{1P} \\ 1 \end{bmatrix} \quad (10.1.21)$$

$$\begin{aligned}
&= \hat{\mathbf{a}}_P' + K_{P+1} \hat{\mathbf{a}}_P^b \\
&= \hat{\mathbf{a}}_P' + K_{P+1} \mathbf{J} \hat{\mathbf{a}}_P'
\end{aligned}$$

where K_{P+1} is a scalar and $\mathbf{J}$ is the permutation matrix

$$\mathbf{J} = \begin{bmatrix} 0 & \cdot & \cdot & \cdot & 0 & 1 \\ 0 & \cdot & \cdot & \cdot & 1 & 0 \\ \cdot & & & & & \cdot \\ \cdot & & & & & \cdot \\ \cdot & \cdot & & & & \cdot \\ 1 & 0 & \cdot & \cdot & \cdot & 0 \end{bmatrix}.$$

The gain K_{P+1} for the backward component $\hat{\mathbf{a}}_P^b = \mathbf{J} \hat{\mathbf{a}}_P'$ is called the *reflection coefficient*. Then, $\hat{\mathbf{a}}_{P+1}$ must satisfy

$$\hat{\mathbf{R}}_{P+1} \hat{\mathbf{a}}_{P+1} = (h_{d0}/a_{0,P+1}) \hat{\delta} \quad (10.1.22)$$

or, substituting from (10.1.21),

$$\hat{\mathbf{R}}_{P+1} \hat{\mathbf{a}}_P' + K_{P+1} \hat{\mathbf{R}}_{P+1} \mathbf{J} \hat{\mathbf{a}}_P' = (h_{d0}/a_{0,P+1}) \hat{\delta}. \quad (10.1.23)$$

Therefore, by the definition of $\hat{\mathbf{a}}_P'$,

$$\begin{bmatrix} -\hat{\mathbf{R}}_P \\ \hat{r}_{P+1} \cdots \hat{r}_1 \end{bmatrix} \hat{\mathbf{a}}_P + K_{P+1} \begin{bmatrix} \hat{r}_1 \cdots \hat{r}_{P+1} \\ \hat{\mathbf{R}}_P \end{bmatrix} \mathbf{J} \hat{\mathbf{a}}_P = \begin{bmatrix} h_{d0}/a_{0,P+1} \\ 0 \\ \cdot \\ \cdot \\ \cdot \\ 0 \end{bmatrix}. \quad (10.1.24)$$

The second through the next-to-last equations in (10.1.24) are satisfied from (10.1.20) because $\hat{\mathbf{R}}_P$ is symmetric and Toeplitz. Letting

$$\Delta_{P+1} = [\hat{r}_{P+1}, \dots, \hat{r}_1] \hat{\mathbf{a}}_P, \quad (10.1.25)$$

the last equation in (10.1.24) gives the reflection coefficient as

$$K_{P+1} = -\Delta_{P+1}/(h_{d0}/a_{0P}). \quad (10.1.26)$$

The first equation, on the other hand, yields from (10.1.20), (10.1.25), and (10.1.26)

$$\begin{aligned}(h_{d0}/a_{0,P+1}) &= (h_{d0}/a_{0P}) + K_{P+1}\Delta_{P+1} \\ &= (h_{d0}/a_{0P}) + K_{P+1}[-K_{P+1}(h_{d0}/a_{0P})] \\ &= (h_{d0}/a_{0P})[1 - K_{P+1}^2].\end{aligned}\quad (10.1.27)$$

Therefore,

$$h_{d0}^2 \leq (h_{d0}/a_{0,P+1}) \leq (h_{d0}/a_{0P}) \quad (10.1.28)$$

because, in the limit as $P \rightarrow \infty$, the impulse response is matched exactly by the AR model, and thus $a_{0\infty} = 1/h_{d0}$. Note, however, that $a_{0N} \neq 1/h_{d0}$ unless $K_P = 0$ for all $P > N$, corresponding to an exact N th-order AR model.

Another important observation may be made concerning the reflection coefficients. From (10.1.28), $(h_{d0}/a_{0P}) > 0$, and thus from (10.1.27)

$$|K_P| < 1, \quad \text{for all } P. \quad (10.1.29)$$

This condition on the reflection coefficients is both necessary and sufficient to ensure the stability of $H(z) = 1/A(z)$, which may be proved by applying Rouché's Theorem [22] to (10.1.21). Rouché's Theorem states that if $F(z)$ and $G(z)$ are analytic functions inside and on a closed contour C (e.g., the unit circle), with $|G(z)| < |F(z)|$ on C , then $F(z)$ and $F(z) + G(z)$ have no zeros on C and the same number of zeros inside C . Let

$$F(z) = z^{P+1}A_P^f(z) = z^{P+1} + \hat{a}_{1P}z^P + \cdots + \hat{a}_{PP}z,$$

which is analytic everywhere except at $z = \infty$. The $P+1$ zeros of $F(z)$ are those of $A_P(z)$ plus one at $z = 0$. Assume that these zeros are all inside the unit circle. Similarly, let

$$\begin{aligned}G(z) &= K_{P+1}z^{P+1}A_P^b(z) = K_{P+1}[\hat{a}_{PP}z^P + \cdots + \hat{a}_{1P}z + 1] \\ &= K_{P+1}A_P^f(z^{-1}).\end{aligned}$$

Hence, the zeros of $G(z)$ are all outside the unit circle (i.e., $G(z)$ is *maximum phase*); but since $|K_{P+1}| < 1$, $|G(z)| < |F(z)|$ for

$z = e^{j\omega}$, that is, on the unit circle. Therefore, by (10.1.21) and Rouché's Theorem, all $P+1$ zeros of $A_{P+1}(z) = z^{-(P+1)}[F(z) + G(z)]$ are inside the unit circle. The sufficiency of the condition $|K_P| < 1$ for stability is thus proved by induction with an initial polynomial $A_0(z) = 1$. The proof of its necessity is similar.

Summary

We will summarize the Levinson–Durbin algorithm by detailing several iterations.

Iteration 0:

The zeroth-order solution is simply the scalar

$$\hat{a}_0 = 1.$$

The normal equation (10.1.20) is $\hat{r}_0 \hat{a}_0 = (h_{d0}/a_{00})$, from which

$$(h_{d0}/a_{00}) = \hat{r}_0.$$

Iteration 1:

The first-order solution is then, from (10.1.21) with $P=0$,

$$\hat{a}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} + K_1 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ K_1 \end{bmatrix} = \begin{bmatrix} 1 \\ \hat{a}_{11} \end{bmatrix}.$$

To determine K_1 , we calculate Δ_1 from (10.1.25) as

$$\Delta_1 = [\hat{r}_1] \hat{a}_0 = \hat{r}_1,$$

and hence from (10.1.26)

$$K_1 = -\Delta_1 / (h_{d0}/a_{00}) = -\hat{r}_1 / \hat{r}_0.$$

Note that $|K_1| < 1$, as expected. Finally, from (10.1.27),

$$\begin{aligned} (h_{d0}/a_{01}) &= (h_{d0}/a_{00})(1 - K_1^2) \\ &= \hat{r}_0(1 - \hat{r}_1^2/\hat{r}_0^2). \end{aligned}$$

Iteration 2:

The second-order solution is then

$$\hat{\mathbf{a}}_2 = \begin{bmatrix} 1 \\ \hat{a}_{11} \\ 0 \end{bmatrix} + K_2 \begin{bmatrix} 0 \\ \hat{a}_{11} \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ K_2(1 + K_2) \\ K_2 \end{bmatrix} = \begin{bmatrix} 1 \\ \hat{a}_{12} \\ \hat{a}_{22} \end{bmatrix}$$

where

$$\begin{aligned} \Delta_2 &= [\hat{r}_2, \hat{r}_1] \hat{\mathbf{a}}_1 = \hat{r}_2 + \hat{a}_{11} \hat{r}_1, \\ K_2 &= -\Delta_2 / (h_{d0}/a_{01}), \\ (h_{d0}/a_{20}) &= (h_{d0}/a_{01})(1 - K_2^2). \end{aligned}$$

The recursion proceeds in this manner through the N th iteration where $P+1=N$, or until $K_{P+1}=0$ which indicates that the AR model exactly matches the given autocorrelation values $\hat{r}(n)$ for $n=0, 1, \dots, P$.

A significant feature of the Levinson–Durbin algorithm is the associated reduction in computation required to solve the normal equations for $\hat{\mathbf{a}}$. The solution of a general set of N linear equations by, for example, Gaussian elimination requires $O(N^3)$ operations and $O(N^2)$ memory locations. Cholesky decomposition requires fewer computations and less memory, but these are still $O(N^3)$ and $O(N^2)$, respectively. In contrast, the Levinson–Durbin algorithm requires only $O(N^2)$ operations and $O(N)$ memory locations. However, in estimating computational complexity, we must also keep in mind that $O(NL)$ computations are needed to calculate the correlation sequence $\hat{r}(m)$, and for $L \gg N$, this dominates the other computations.

Step-Down Algorithm

The reflection coefficient sequence $\{K_1, K_2, \dots, K_N\}$ is equivalent to the vector $\hat{\mathbf{a}}$ in the sense that one can be obtained directly from the other. The *step-up* algorithm to convert $\{K_P\}$ to $\hat{\mathbf{a}}$ is simply the recursive application of (10.1.21) with $\hat{\mathbf{a}}'_0 = [1 \ 0]^T$ to obtain $\hat{\mathbf{a}}_P$, $P=1, 2, \dots, N$. The *step-down* algorithm converts $\hat{\mathbf{a}}$ to $\{K_P\}$ and is also derived from (10.1.21). Those equations may be grouped into pairs of the form

$$\begin{aligned} \hat{a}_{nP} &= \hat{a}_{n,P-1} + K_P \hat{a}_{P-n,P-1}, \\ \hat{a}_{P-n,P} &= \hat{a}_{P-n,P-1} + K_P \hat{a}_{n,P-1}, \end{aligned} \tag{10.1.30}$$

which may be solved for $\hat{a}_{n,P-1}$ to produce

$$\hat{a}_{n,P-1} = \frac{\hat{a}_{nP} - K_P \hat{a}_{P-n,P}}{1 - K_P^2}. \quad (10.1.31)$$

Also, from the last equation in (10.1.21), we have $K_P = \hat{a}_{PP}$. Hence, $\hat{a}_{P-1}$ may be derived from $\hat{a}_P$ recursively for $P = N, N-1, \dots, 2$ to produce $\{K_P\}$. Note that for the autocorrelation method, $|K_P| < 1$, and thus the denominator $1 - K_P^2$ in (10.1.31) cannot be zero. However, in other applications of the step-down algorithm, we can have $|K_P| = 1$, in which case (10.1.31) cannot be used.

An additional application of the step-down algorithm is that, given an arbitrary $\hat{a}$, it constitutes a simple stability test for the corresponding $H(z) = 1/A(z)$. That is, if $|K_P| \geq 1$ for any P , then $H(z)$ is not stable. For example, although the covariance method does not guarantee stability, the resulting model can be easily checked for stability by means of the step-down algorithm.

Magnitude Response Data

In many applications, one is not given a causal sequence $h_d(n)$ to approximate, but rather $|H'_d(\omega)|$, from which $\hat{r}(m)$ must be estimated. Sampling $|H'_d(\omega)|$ on a dense set of uniformly spaced frequencies $\omega_k = 2\pi k/K$, we produce the DFT $|\hat{H}_d(k)|$, which may be inverted to yield $\hat{h}_d(n)$. However, $|H'_d(\omega)|$ is zero-phase, and thus $\hat{h}_d(n)$ is noncausal and symmetric about $n = 0$. Therefore, let

$$\hat{r}(m) = \sum_{n=-L}^{L-m} \hat{h}_d(n) \hat{h}_d(n+m) \quad (10.1.32)$$

for $m \geq 0$ and $L < K/2$.

As an example, we apply the autocorrelation method to approximate the frequency response $|H'_d(\omega)|$ shown in figure 10.2(a) with $K = 500$, $L = 50$, and $N = 20$. The resulting AR design $|H'(\omega)| = 1/|A'(\omega)|$ is shown in figure 10.2(b). Note that the greatest deviation from the desired response occurs at the bottom of the V-shaped notch. This is to be expected since this is an all-pole design and we have no zeros with which to produce a sharp notch. The passbands are almost equiripple, but this is not a general property of AR designs. In fact, in the vicinity of abrupt transitions, large ripples will be produced; and hence such abrupt changes are to be avoided in the specification of $|H'_d(\omega)|$.

Steiglitz- McBride Method

The linear normal equations for the coefficient vector $\underline{a}$ in this section have resulted from the choice of the inverse-filter error $e(n) = h_d(n) * a(n) - \delta(n)$ in (10.1.5) to which to apply the least-

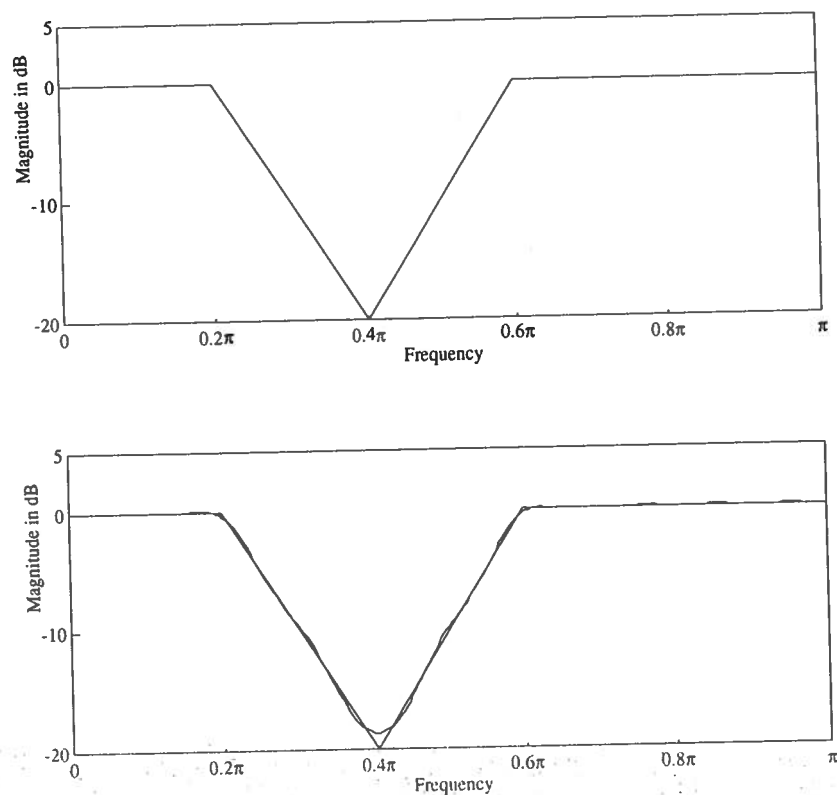


Figure 10.2. (a) Desired magnitude response and (b) 20th-order AR design.

squares error criterion. It may be argued [23] that this is not the error we really wish to minimize and hence that the resulting estimate $\hat{a}$ is not the best least-squares estimate. Rather, by these arguments, the fitting error

$$e(n) = h_d(n) - h(n) \quad (10.1.33)$$

is the preferred error measure. The least-squares norm E_N then becomes

$$E_N = \sum_{n=0}^L [h_d(n) - h(n)]^2 \quad (10.1.34)$$

or, in the limiting case as $L \rightarrow \infty$,

$$\begin{aligned} E_N &= \sum_{n=0}^{\infty} [h_d(n) - h(n)]^2 \\ &= \frac{1}{2\pi} \int_0^{2\pi} |H'_d(\omega) - H'(\omega)|^2 d\omega. \end{aligned} \quad (10.1.35)$$

This error norm, however, leads to nonlinear equations for $\underline{a}$, which must be solved iteratively. Steiglitz and McBride [24] have proposed an efficient iterative algorithm to estimate both the denominator and numerator coefficients for $H(z)$ in the more general ARMA case, and we will adapt their algorithm here to the AR-only case.

We note first that

$$\begin{aligned} h_d(n) &= h_d(n) * \delta(n) = h_d(n) * [h(n) * a(n)] \\ &= h(n) * h_d(n) * a(n). \end{aligned}$$

Therefore, defining

$$\mathbf{H} = \begin{bmatrix} h_0 & 0 & \cdot & \cdot & \cdot & 0 \\ h_1 & h_0 & & & & 0 \\ \cdot & & \cdot & & & \cdot \\ \cdot & & & \cdot & & \cdot \\ \cdot & & & & \cdot & \cdot \\ h_L & \cdot & \cdot & \cdot & \cdot & h_0 \end{bmatrix}, \quad \underline{h}_d = \begin{bmatrix} h_{d0} \\ h_{d1} \\ \cdot \\ \cdot \\ \cdot \\ h_{dL} \end{bmatrix}, \quad \underline{h} = \begin{bmatrix} h_0 \\ h_1 \\ \cdot \\ \cdot \\ \cdot \\ h_L \end{bmatrix},$$

we have in vector form from (10.1.33)

$$\underline{h}_d = \mathbf{H}\mathbf{H}_d \underline{a} = \underline{h} + \underline{e}. \quad (10.1.36)$$

The minimizing solution $\underline{a}$ is approximately

$$\begin{aligned} \underline{a} &= (\mathbf{H}\mathbf{H}_d)^{\#} \underline{h} \\ &= (\mathbf{H}'_d \mathbf{H}' \mathbf{H} \mathbf{H}_d)^{-1} (\mathbf{H}' \mathbf{H}_d)' \underline{h}, \end{aligned} \quad (10.1.37)$$

where $()^{\#}$ denotes the pseudo-inverse matrix. This solution is only approximate (although quite close to the exact solution) because we have neglected the fact that the model $h(n)$ and thus $\mathbf{H}$ and $\underline{h}$

are themselves functions of $\underline{a}$. The matrix product $\mathbf{H}\mathbf{H}_d$ corresponds to the convolution $h(n) * h_d(n)$. Hence, (10.1.36) tells us to prefilter the data $h_d(n)$ by the model $h(n)$ before solving for $\underline{a}$, which we can only do iteratively. That is, letting the i th estimate be $\underline{a}_i$, with corresponding $\mathbf{H}_i$ and $\underline{h}_i$ we solve for $\underline{a}_{i+1}$ from (10.1.37) by

$$\underline{a}_{i+1} = (\mathbf{H}_i \mathbf{H}_d)^{\#} \underline{h}_i. \quad (10.1.38)$$

The convergence of this iterative algorithm has been found to be rapid (on the order of 5 to 10 iterations) [23]. Either the covariance or autocorrelation method can be used to provide the initial vector $\underline{a}_0$. A potential difficulty is that the algorithm may not converge if the order (N) of the model $h(n)$ is chosen to be larger than that of the data $h_d(n)$. Hence, it is important not to overestimate the order of the model.

†10.2 / Moving-Average (All-Zero) Filters

Moving-average (MA) filters are another name for FIR filters. From chapter 9, we already have three techniques for designing linear-phase FIR filters with arbitrary magnitude responses. Another class of techniques that generally provides minimum-phase filter designs is available from the field of statistical modeling. Two of these techniques are described in this section.

Durbin's Method

The first technique is Durbin's method [25], which is straightforward and highly effective if an MA model is appropriate to the data. The MA filter is of the form $H(z) = B(z)$, where

$$B(z) = b_0 + b_1 z^{-1} + \cdots + b_M z^{-M}. \quad (10.2.1)$$

Since an L th-order AR model can approximate any desired filter with arbitrary accuracy if L is made sufficiently large (as discussed in the previous section), let $1/A_L(z)$ be a high-order AR model of $h_d(n)$ or $|H'_d(\omega)|$ with $L \gg M$. As noted previously, the AR filter cannot directly model sharp notches in $|H'_d(\omega)|$, but for sufficiently large L , the "information" concerning such notches is nevertheless contained in $A_L(z)$ and can be extracted in the design of $B(z)$.

Therefore, we desire

$$B(z) \approx 1/A_L(z) \quad (10.2.2)$$

or, in the time domain,

$$a_L(n) * b(n) = \delta(n) + e(n), \quad (10.2.3)$$

where $e(n)$ is the approximation error. This is exactly analogous to (10.1.5), and thus we estimate $b(n)$ such that $1/B(z)$ is an AR model of the FIR filter $A_L(z)$. Since $A_L(z)$ is FIR, the autocorrelation method may be directly applied to the "data" sequence $a_L(n)$ (without windowing) to compute $b(n)$. Therefore, $B(z)$ is guaranteed to be minimum phase, whether the polynomial $A_L(z)$ is or not.

EXAMPLE

To find a simple first-order MA model by Durbin's method, let the filter $H_d(z)$ to be modeled be the first-order AR filter

$$H_d(z) = \frac{1}{1 - az^{-1}}.$$

Clearly then, $A_L(z)$ equals $1 - az^{-1}$ exactly in this case. The normal equation for the autocorrelation method applied to the coefficient sequence $a_L(n) = \delta(n) - a\delta(n-1)$ is simply

$$\begin{bmatrix} 1 + a^2 & -a \\ -a & 1 + a^2 \end{bmatrix} \begin{bmatrix} \hat{b}_0 \\ \hat{b}_1 \end{bmatrix} = \begin{bmatrix} 1/b_0 \\ 0 \end{bmatrix},$$

from which

$$\hat{b}_1 = \frac{a}{1 + a^2}.$$

We note that $\hat{b}_1 \approx a$ for $a \ll 1$, as expected, because $h_d(n) = a^n u(n)$ and the terms for $n \geq 2$ are negligible if $a \ll 1$.

As another example, an MA model with $M = 20$ was made of the ideal filter in figure 10.2(a) using an intermediate AR model of order $L = 128$. Such a high-order AR model is not difficult to compute via the autocorrelation method with the Levinson-Durbin algorithm. The resulting magnitude response is shown in figure 10.3. Comparing this with the AR model having $N = 20$ in figure 10.2(b), we note that the notch for the MA design is deeper, as expected; but the accuracy near the band edges is not as good as

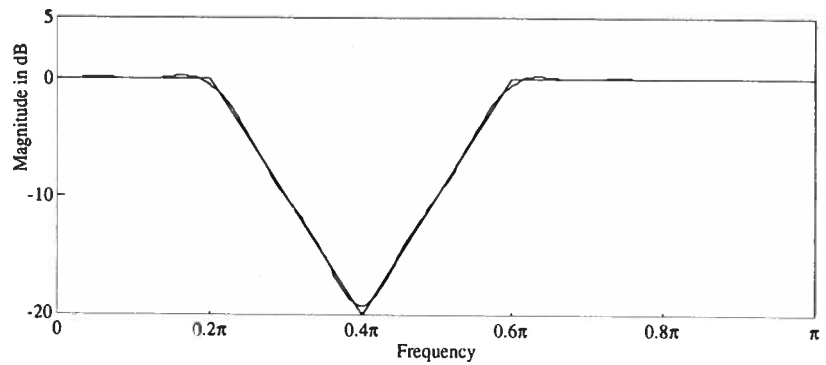


Figure 10.3. Twentieth-order MA design from Durbin's method for specification in figure 10.2.

in the AR case because we now have no poles with which to produce sharp peaks or edges in $|H'(\omega)|$. The ripple in the pass-bands of both designs is about the same.

Durbin's method has several other useful applications besides direct estimation of MA models. Given a polynomial $B(z)$ that is not minimum phase (such as the denominator $A(z)$ of an unstable AR filter), a minimum-phase equivalent $\tilde{B}(z)$ with $|\tilde{B}'(\omega)| \approx |B'(\omega)|$ can be obtained by factoring the zero-phase polynomial $G(z) = B(z)B(z^{-1})$. That is, forming the autocorrelation function

$$g(n) = b(n) * b(-n), \quad (10.2.4)$$

the Yule-Walker equation is solved to produce an AR model $A_L(z)$ with $L \gg M$ (noting that $g(n) = 0$ for $|n| > M$). Alternatively, we may be given $G(z)$ itself to factor. If $G(z)$ is factorable, then Durbin's method provides a close approximation to the minimum-phase factor. If, however, $G(z)$ is not factorable (corresponding to single zeros on the unit circle and thus a nonpositive power spectrum $G'(\omega)$), the estimate $\tilde{B}(z)$ cannot be a true factor. Nevertheless, it can be a useful approximate factor in the sense that $\tilde{G}(z) = \tilde{B}(z)\tilde{B}(z^{-1})$ is now factorable and $\tilde{G}'(\omega)$ is positive. For example, figure 10.4 shows an MA power spectrum $G'(\omega)$ with negative values (off the dB scale) for two intervals on the ω axis, along with $\tilde{G}'(\omega)$ from Durbin's method. Note that $\tilde{G}'(\omega)$ closely

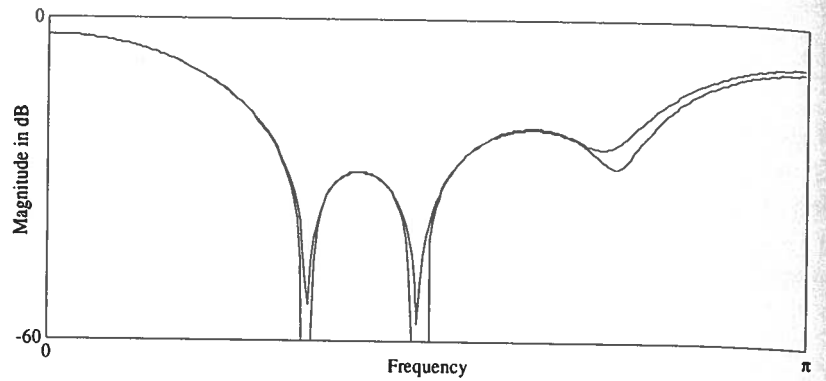


Figure 10.4. Unfactorable $G'(\omega)$ versus factorable $\tilde{G}'(\omega)$ from Durbin's method.

approximates $G'(\omega)$ everywhere except in those intervals where $G'(\omega)$ is negative.

Judell's Method

Each of the two steps in Durbin's method employs a whitening criterion to design the inverse filters $A_L(z)$ and $B(z)$. A single, overall whitening criterion may be applied by requiring that $H_d(z)/B(z)$ be approximately white. That is, defining the inverse filter

$$H_b(z) = 1/B(z), \quad (10.2.5)$$

the equivalent of (10.1.5) in the MA case is

$$h_d(n) * h_b(n) = \delta(n) + e(n). \quad (10.2.6)$$

Minimization of the least-squares norm of $e(n)$, however, now leads to nonlinear equations for $b(n)$. An iterative algorithm, similar to that of Steiglitz and McBride, has been proposed by Judell [26] to estimate $b(n)$. Hannan [80] has proposed a similar method in terms of DFTs. Noting that

$$h_b(n) * h_b(n) * h_d(n) * b(n) = \delta(n) + e(n) \quad (10.2.7)$$

and letting

$$\mathbf{H}_b = \begin{bmatrix} h_{b0} & 0 & \cdot & \cdot & \cdot & 0 \\ h_{b1} & h_{b0} & & & & 0 \\ \cdot & & \cdot & & & \cdot \\ \cdot & & & \cdot & & \cdot \\ \cdot & & & & \cdot & \cdot \\ h_{bL} & \cdot & \cdot & \cdot & \cdot & h_{b0} \end{bmatrix}, \quad (10.2.8)$$

we have

$$\mathbf{H}_b \mathbf{H}_b \mathbf{H}_d \underline{b} = \underline{\delta} + \underline{e}. \quad (10.2.9)$$

The minimizing solution $\underline{b}$ is thus approximately

$$\underline{b} = (\mathbf{H}_b \mathbf{H}_b \mathbf{H}_d)^{\#} \underline{\delta}, \quad (10.2.10)$$

and the implied iteration is

$$\underline{b}_{i+1} = (\mathbf{H}_{bi} \mathbf{H}_{bi} \mathbf{H}_d)^{\#} \underline{\delta}. \quad (10.2.11)$$

The solution from Durbin's method makes an excellent starting vector $\underline{b}_0$ for the iteration. An unexpected difficulty is that the iteration given by (10.2.11) goes in exactly the wrong direction. That is, letting $\Delta \underline{b}_{i+1} = \underline{b}_{i+1} - \underline{b}_i$, the elements of $\Delta \underline{b}_{i+1}$ have appropriate magnitudes, but incorrect signs [26]. Therefore, we define a modified iteration by

$$\begin{aligned} \hat{\underline{b}}_{i+1} &= \hat{\underline{b}}_i - \Delta \underline{b}_{i+1} \\ &= 2\hat{\underline{b}}_i - \underline{b}_{i+1}. \end{aligned}$$

The modified iteration converges rapidly (3 to 5 iterations).

As a final note, we observe that the fitting error $e(n) = h_d(n) - h(n)$, with $h(n) = b(n)$, is not usually an appropriate error measure for the least-squares criterion in the MA case, unlike the AR and ARMA cases, because the minimizing solution is then just the rectangularly windowed response $b(n) = h_d(n)w_R(n)$.

†10.3 / ARMA (Pole/Zero) Filters

The general ARMA case is, as might be expected, more complicated than the separate AR and MA cases previously investigated.